MACHINE LEARNING IN PHYSICS
FOUNDATIONS 2

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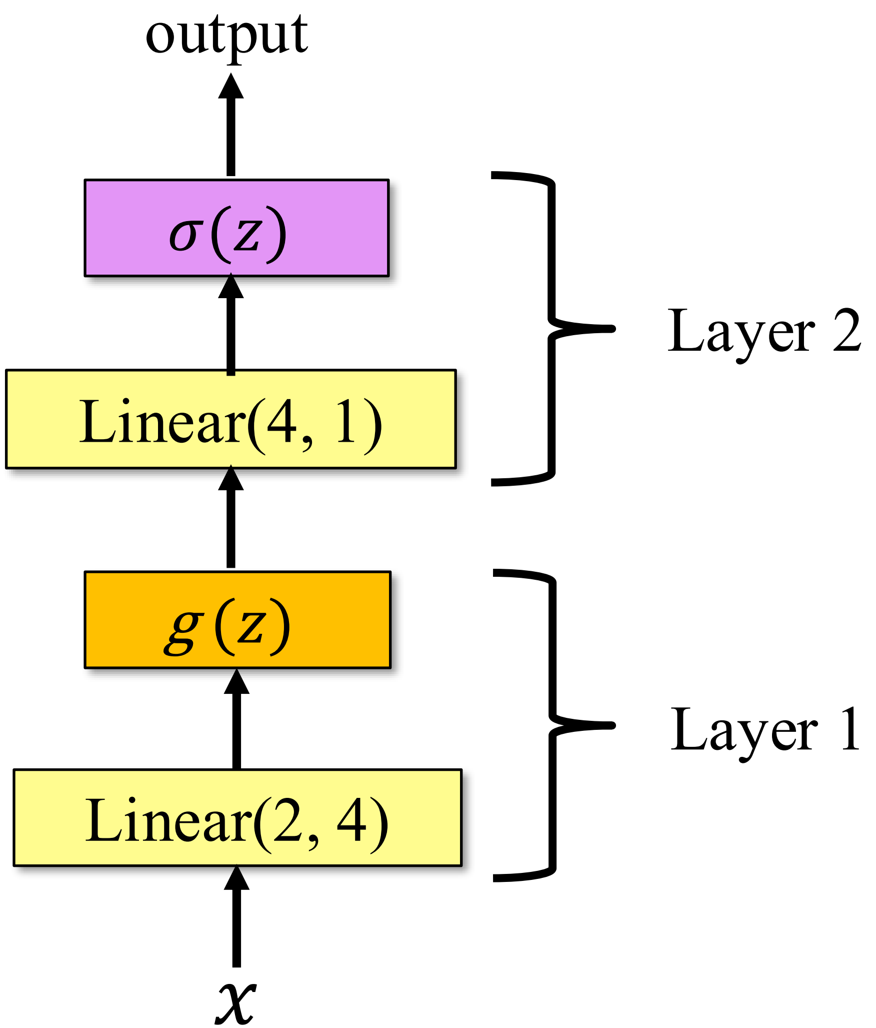
Recap

Last time we introduced the multi-node, multi-layer perceptron (MLP). This is also known as a **feed-forward neural network** (FCNN)



$\sigma(z) = (1 + e^{-z})^{-1}$

$g(z) = \mathrm{relu}(z)$

$f(z) = zA^T + b$

FITTING MODELS TO DATA

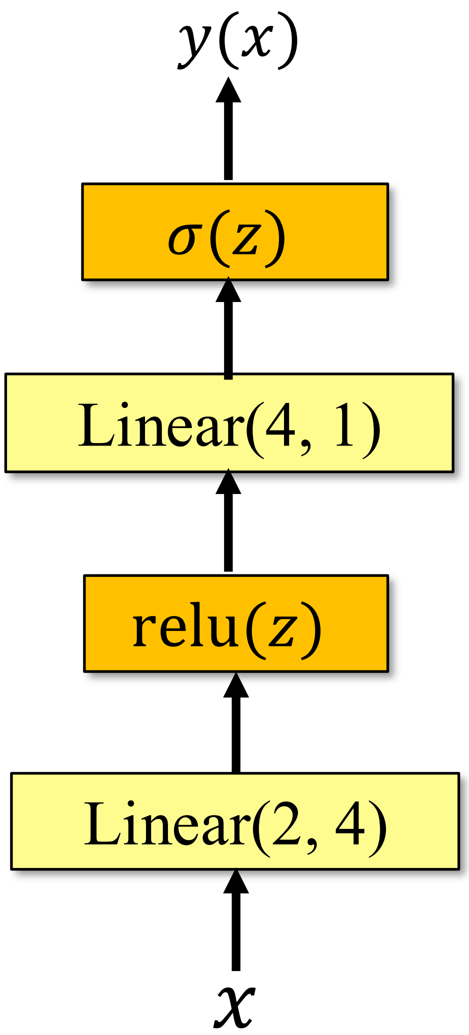
Fitting Models to Data (1)

Suppose our goal is to approximate an unknown function

$$y = f(x_1, x_2)$$

given a dataset $D = \{(x_1, x_2, y)_{i=1}^{N}\}$.

The standard approach is to approximate $f(x_1, x_2)$ with a *parameterized* function $f_\omega(x_1, x_2)$, that is, a model, where ω are the parameters of the model.



Fitting Models to Data (2)

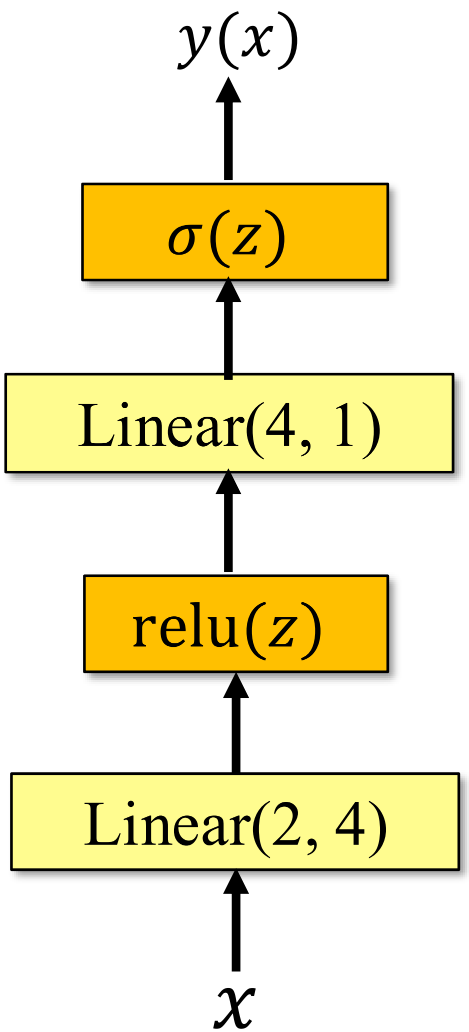
One way to find the *best-fit values* ω^* of the model $f(x_1, x_2, \omega)$ is to minimize the **quadratic form**

$$R(\omega) = \frac{1}{N}\sum_{i=1}^{N}(y_i - f_i)^2$$

with respect to the parameters ω, where

$$f_i \equiv f_\omega(x_{1i}, x_{2i}).$$

This is called a **least-squares fit**.



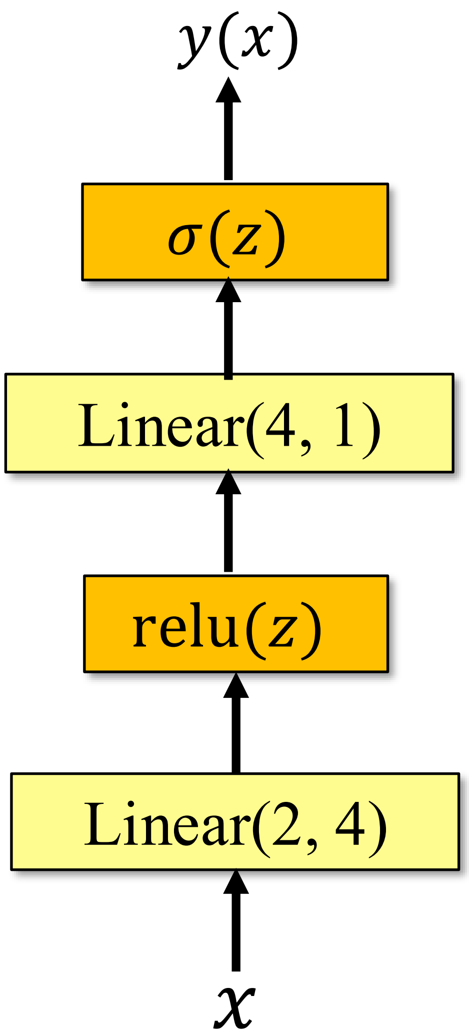
Fitting Models to Data (3)

The least-squares fit can be generalized to

$$R(\omega) = \frac{1}{N}\sum_{i=1}^{N} L(y_i, f_i)$$

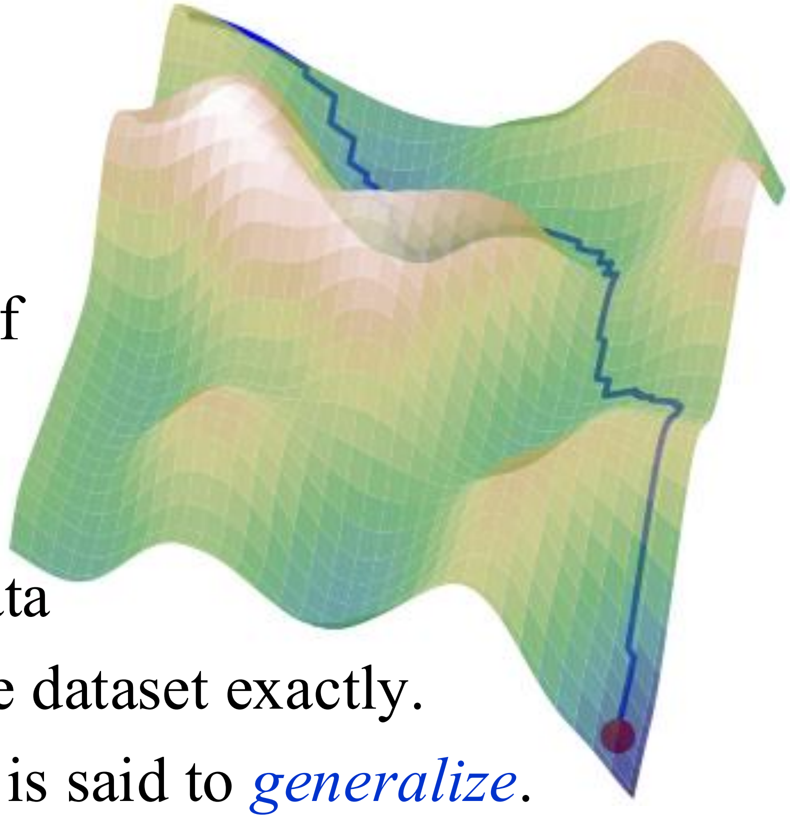
where, $L(y_i, f_i)$, the **loss function**, measures the discrepancy between the desired **target** y and the model, f.

$R(\omega)$ is called the **empirical risk** or **average loss**. **Warning:** In ML, the empirical risk is often referred to as the "loss".



Empirical Risk Landscape (1)

The empirical risk defines a highly corrugated high-dimensional "landscape" in the space of parameters.

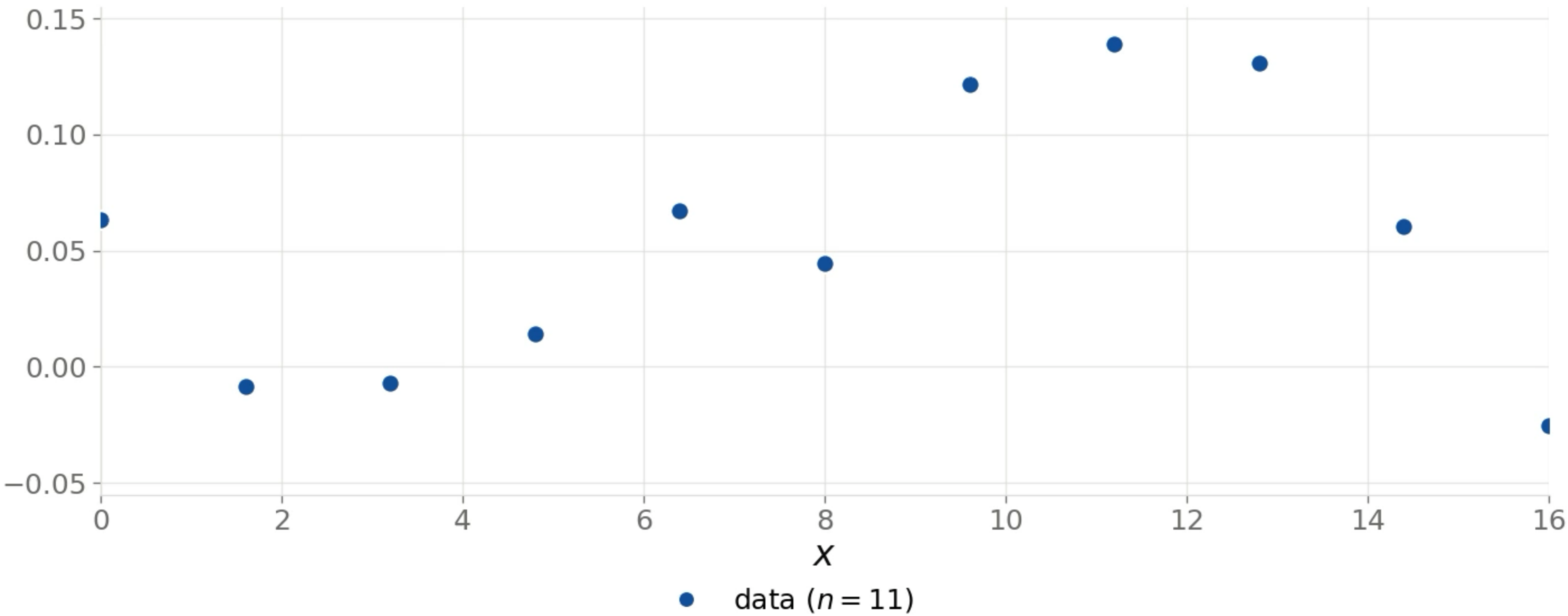


Our task is to navigate the empirical risk landscape to find a *good approximation* of the minimum not of *that* landscape but rather of the landscape that would be formed if we if we had an *infinite* amount of data and a model that would fit the infinite dataset exactly. A model that approximates this ideal is said to *generalize*.

Empirical risk is what we can measure — generalization is what we want

We only ever see a finite, noisy sample.

Everything we compute is built from these 11 points.



Example: A least squares fit to a **linear model**.

$$f(x, \omega) = \sum_{j=0}^{n} \omega_j g_j(x)\,, \quad g_j(x) = x^j$$

RISK FUNCTIONAL

Risk Functional (1)

In mathematics and physics, insight can often be gained by taking some limit of an expression. Consider the limit of

$$R(\omega) = \frac{1}{N} \sum_{i=1}^{N} L(y_i, f_i)$$

as $N \to \infty$, that is, as the amount of data grows without limit.

In that limit, the empirical risk becomes the **risk functional**,

$$R[f] = \int dx \int dy \, L(y, f) \, p(x, y)$$

where $p(x, y) dx dy$ is the probability distribution of the data.

Risk Functional (2)

To find the optimal function $f = f^*$, ideally, we should minimize the risk functional

$$R[f] = \int dx \int dy \, L(y, f) \, p(x, y)$$

While we know the functional form of the loss function $L(y, f)$ because we choose it, the probability distribution, $p(x, y) dx dy$, of the data is usually unknown.

However, because $R(\omega)$ is a **Monte Carlo** approximation of $R[f]$, any useful result derived for the risk functional pertains, at least approximately, to the empirical risk.

Risk Functional (3)

To minimize

$$R[f] = \int dx \int dy\, L(y, f)\, p(x, y)$$

first note that $p(x, y) = p(y|x)\, p(x)$.

Therefore, we can write the functional $R[f]$ as

$$R[f] = \int dx\ \ p(x)\, \boldsymbol{\mathcal{L}(x, f)}$$

where,

$$\boldsymbol{\mathcal{L}(x, f)} = \int dy\, L(y, f)\, p(y|\, x)$$

Risk Functional (4)

Add an *arbitrary* function $\epsilon g(x)$ to the optimal function f^*, where ϵ is an infinitesimal*.

$$R[f^* + \epsilon g] = \int dx\ \ p(x)\ \boldsymbol{\mathcal{L}}(\boldsymbol{x}, \boldsymbol{f}^* + \boldsymbol{\epsilon g})$$

$$= \int dx\ \ p(x)\left(\boldsymbol{\mathcal{L}}(\boldsymbol{x}, \boldsymbol{f}^*) + \boldsymbol{\epsilon g}\frac{\partial \boldsymbol{\mathcal{L}}}{\partial f^*}\right)$$

$$= R[f^*] + \boldsymbol{\epsilon}\int dx\, p(x)\boldsymbol{g}(\boldsymbol{x})\frac{\partial \boldsymbol{\mathcal{L}}}{\partial f^*}$$

* $0 < |\epsilon| < |x|$, where x is any real number.

Risk Functional (5)

Rearranging, and taking standard part* of both sides, we find

$$\mathrm{st}\left(\frac{R[f^* + \epsilon g] - R[f^*]}{\epsilon}\right) = \int dx\, p(x) g(x) \frac{\partial \mathcal{L}}{\partial f^*}$$

The lefthand side defines the **functional derivative** $\delta R / \delta f$.

By assumption, $\delta R / \delta f$ is zero at $f = f^*,\ \forall\, x$. Therefore,

$$\int dx\, p(x) g(x) \frac{\partial \mathcal{L}}{\partial f} = 0$$

when $f = f^*$.

*The standard part of the hyperreal $x^* = x + \epsilon a$ is defined by $x = \mathrm{st}(x^*)$

Risk Functional (6)

We want the expression

$$\int dx\, p(x) g(x) \frac{\partial \mathcal{L}}{\partial f} = 0$$

to hold for *any* function $g(x)$ and $\forall\, x$.

This can happen *if only if*

$$\frac{\partial \mathcal{L}}{\partial f} = 0$$

Assuming the integral and partial derivative operations commute, and recalling that $\mathcal{L}(x, f) = \int dy\, L(y, f)\, p(y|\, x)$,

we arrive at the

Very **I**mportant **R**esult:

$$\boxed{\int dy\, \frac{\partial L}{\partial f} p(y \mid x) = 0}$$

Risk Functional (7)

Points to Note:

$$\boxed{\int dy\, \frac{\partial L}{\partial f} p(y \mid x) = 0}$$

1. This result is independent of the details of $f_\omega(x)$. However,…
2. The function $f_\omega(x)$ must have sufficient **capacity**: i.e., there must exist an approximation $f_{\widehat{\omega}}(x)$ found by minimizing the empirical risk $R(\omega)$ that is arbitrarily close to the optimal function f^*.
3. Moreover, it must be possible to find that function!

COMMON LOSS FUNCTIONS

Common Loss Functions (1)

Quadratic loss (often used for regression)

$$L(y,f) = (y-f)^2$$

Binary cross entropy (often used for classification)

$$L(y,f) = -[y\log f + (1-y)\log(1-f)]$$

Exponential loss

$$L(y,f) = \exp(-wyf/2)$$

Quantile loss ($0 \le \tau \le 1$)

$$L(y,f) = \begin{cases} \tau(y-f) & y \ge f \\ (1-\tau)(f-y) & y < f \end{cases}$$

Zero-One loss

$$L(y,f) = \begin{cases} 0 & |y-f| \le b \\ 1 & |y-f| > b \end{cases}$$

Common Loss Functions (2)

Quadratic loss: $L(y, f) = (y - f)^2$

$$\int \frac{\partial L}{\partial f}\, p(y|x)\, dy = 0$$

Solution

$$\boxed{f(x, \omega^*) = \int y\, p(y \mid x)\, dy}$$

Very Important Point (VIP): As noted, the result is independent of the details of f. The result depends solely on the form of the loss function and the conditional probability, $p(y|x)dy$, of the data.

Binary Cross Entropy Loss (1)

Binary cross entropy loss : $-[y\log f + (1-y)\log(1-f)]$

$$\int \frac{\partial L}{\partial f}\, p(y|x)\, dy = 0$$

$$\int \left[\frac{y}{f} - \frac{(1-y)}{1-f}\right] p(y|x)\, dy = 0$$

Note: $y = \{0, 1\}$. Therefore,

$$-\frac{p(y=0|x)}{1-f} + \frac{p(y=1|x)}{f} = 0,$$

which implies

$$\boxed{f(x,\omega^*) = p(y=1|x)}$$

Binary Cross Entropy Loss (2)

The result

$$\boxed{f(x,\omega^*) = p(1|x) = \frac{p(x|1)\,\pi(1)}{p(x)}}$$

with $p(x) = p(x|1\,)\,\pi(1) + p(x|0\,)\,\pi(0),$ $(k \equiv y = k)$, shows that *any* model trained using the the cross-entropy loss necessarily approximates a **class probability**, that is, the probability that an object characterized by the **features** (i.e., attributes) x belongs to the class labeled $y = 1$.

- $p(x|k)$ is the probability density of x for class $y = k$.
- $\pi(k)$ is the prior probability of class $y = k$.

Binary Cross Entropy Loss (3)

Example: A Spam Filter

A spam filter is a binary classifier: spam vs. non-spam emails.

Suppose $y = 1$ and $y = 0$ labels spam and non-spam emails, respectively, and the training dataset comprises 50% spam emails and 50% non-spam emails. That is, the dataset is **balanced**.

- Each email is characterized by a vector x, and $p(x|1)$ and $p(x|0)$ are the probability densities of these vectors for spam and non-spam emails, respectively.

- The prior probabilities are $\pi(1) = 0.50$ and $\pi(0) = 0.50$.

Summary

Stochastic Gradient Descent

- Many (1st-order) minimization algorithms are variations of the following algorithm:

$$\omega_{j+1} = \omega_j - \eta \nabla R$$

- Batches of data are used, which introduces noise into ∇R.

Risk Minimization

- Given a data set $D = \{(x, y)\}_{i=1}^{N}$, a model $f(x, \omega)$, and a loss function $L(y, f)$, the optimal function $f^* = f(x, \omega^*)$ satisfies:

$$\int \frac{\partial L}{\partial f} p(y \mid x) dy = 0$$